

Real Estate Agency Portal

CIS344 — Spring 2026

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1. Project Purpose

The Real Estate Agency Portal is a web-based platform built with PHP and MySQL that connects agents, buyers, and renters in a realistic agency environment. The system allows agents to manage property listings, while buyers and renters can browse available properties, save favorites, and submit inquiries.

2. Database Design

The database is named `real_estate_portal_db` and contains five tables, each representing a core entity in the agency system.

2.1 Tables Overview

Table	Primary Key	Description
Users	userId	Stores all portal users: agents, buyers, and renters
Properties	propertyId	Stores real estate listings with agent assignment
Inquiries	inquiryId	Records messages sent by buyers/renters about properties
Transactions	transactionId	Records completed sales or rentals
Favorites	favoriteId	Stores properties saved by buyers and renters

2.2 Users Table

Stores every user of the portal regardless of role.

Column	Type	Notes
userId	INT AUTO_INCREMENT	Primary Key
userName	VARCHAR(50)	Unique, required
contactInfo	VARCHAR(200)	Email or phone
passwordHash	VARCHAR(255)	Bcrypt hash via <code>password_hash()</code>
userType	ENUM	agent, buyer, or renter

2.3 Properties Table

Stores all property listings. Each listing is owned by an agent via a foreign key.

Column	Type	Notes
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propertyId	INT AUTO_INCREMENT	Primary Key
title	VARCHAR(100)	Listing title
propertyType	VARCHAR(50)	House, Condo, Apartment, etc.
address	VARCHAR(200)	Street address
city	VARCHAR(100)	City name
price	DECIMAL(12,2)	Listing price
status	ENUM	available, sold, or rented
agentId	INT	Foreign Key → Users(userId)

2.4 Inquiries Table

Records inquiry messages submitted by buyers or renters about specific properties.

Column	Type	Notes
inquiryId	INT AUTO_INCREMENT	Primary Key
userId	INT	Foreign Key → Users(userId)
propertyId	INT	Foreign Key → Properties(propertyId)
message	VARCHAR(255)	Inquiry text
inquiryDate	DATETIME	Timestamp of submission

2.5 Transactions Table

Records completed sales or rental agreements between a user and a property.

Column	Type	Notes
transactionId	INT AUTO_INCREMENT	Primary Key
propertyId	INT	Foreign Key → Properties(propertyId)
userId	INT	Foreign Key → Users(userId)
transactionType	ENUM	sale or rental
transactionDate	DATETIME	Timestamp of transaction
amount	DECIMAL(12,2)	Transaction amount

2.6 Favorites Table

Allows buyers and renters to save properties for later reference.

Column	Type	Notes
favoriteId	INT AUTO_INCREMENT	Primary Key
userId	INT	Foreign Key → Users(userId)
propertyId	INT	Foreign Key → Properties(propertyId)
savedDate	DATETIME	Timestamp when property was saved

2.7 View: PropertyListingView

A database view that joins the Properties and Users tables to display listing details alongside the agent name and contact information, simplifying queries across the application.

2.8 Stored Procedures

AddOrUpdateUser — Accepts an optional userId. If NULL is passed, a new user is inserted. If an ID is provided, the matching record is updated. If no match is found, the user is inserted.

ProcessTransaction — Inserts a transaction record and updates the associated property status to sold or rented depending on the transaction type.

2.9 Trigger: AfterTransactionInsert

Automatically fires after any row is inserted into the Transactions table. If the transaction type is sale, the property status is set to sold. If the type is rental, the status is set to rented. This ensures property status stays consistent even when transactions are inserted manually.

3. PHP Functionality

The application is organized into a clean folder structure with separate directories for configuration, database classes, shared includes, assets, and SQL scripts.

3.1 Folder Structure

Folder / File	Purpose
config/config.php	Database constants and session initialization
classes/Database.php	PDO connection class
classes/RealEstateDatabase.php	All database methods
includes/auth.php	requireLogin() and requireRole() functions
includes/header.php	Shared HTML header and navigation
includes/footer.php	Shared HTML footer
assets/style.css	Site-wide stylesheet
sql/real_estate_portal_db.sql	Complete database schema and sample data

3.2 RealEstateDatabase Class Methods

Method	Purpose
addUser()	Inserts a new user into the Users table using a prepared statement
getUserByUsername()	Retrieves a single user record by username
getUserDetails()	Returns a user with their related inquiries, favorites, and transactions
addProperty()	Inserts a new property listing into the Properties table
getAllProperties()	Returns all listings using PropertyListView
getPropertyById()	Returns one listing by ID using PropertyListView
getPropertiesByCity()	Filters listings by city using PropertyListView
addInquiry()	Inserts a new inquiry with the current timestamp
addFavorite()	Saves a property to a user's favorites
removeFavorite()	Removes a saved property from favorites
getFavoritesByUser()	Returns all favorites for a given user
isFavorite()	Checks whether a property is already in a user's favorites
processTransaction()	Calls the ProcessTransaction stored procedure

3.3 PHP Pages

Page	Access	Description
index.php	Public	Homepage with project overview and role descriptions
register.php	Public	Registration form with password hashing
login.php	Public	Login form with password verification and session creation
logout.php	Logged in	Destroys session and redirects to homepage
dashboard.php	Logged in	Role-specific dashboard for all user types
properties.php	Public	Browse all available property listings
property_details.php	Public	View a single listing with inquiry and favorite options
add_property.php	Agent only	Form to add a new property listing
edit_property.php	Agent only	Pre-filled form to edit an existing listing
submit_inquiry.php	Buyer/Renter	Form to send a message about a property
favorites.php	Buyer/Renter	View and manage saved properties

4. Login and Session Handling

Sessions are initialized in `config/config.php` using `session_start()`, which is included on every page via `header.php`. This ensures session data is available throughout the application.

4.1 Registration

When a user submits the registration form, the system first checks whether the username already exists by calling `getUserByUsername()`. If the username is taken, the user is shown an error. Otherwise, the submitted password is hashed using PHP's `password_hash()` function with the `PASSWORD_DEFAULT` algorithm, which produces a secure bcrypt hash. The hashed value is stored in the Users table — the plain text password is never saved.

4.2 Login

On login, the system retrieves the user record by username and passes the submitted password and the stored hash to `password_verify()`. If the verification succeeds, the user record is stored in the `$_SESSION` superglobal. The session stores the `userId`, `userName`, `contactInfo`, `passwordHash`, and `userType`, which are used throughout the application to personalize the experience and enforce access rules.

4.3 Logout

The `logout.php` page calls `session_unset()` to clear all session variables and `session_destroy()` to terminate the session, then redirects the user to the homepage.

5. Role-Based Access Control

The application supports three user roles: agent, buyer, and renter. Access control is enforced in `includes/auth.php` through two helper functions.

5.1 Helper Functions

`requireLogin()` — Redirects unauthenticated users to the login page if no session is active.

`requireRole(array $roles)` — Calls `requireLogin()` first, then checks whether the logged-in user's type matches one of the allowed roles. If not, the user sees an Access Denied message.

5.2 Role Permissions

Feature	Agent	Buyer	Renter
Browse Properties	Yes	Yes	Yes
View Property Details	Yes	Yes	Yes
Add Property	Yes	No	No
Edit Property	Yes	No	No
Submit Inquiry	No	Yes	Yes
Save to Favorites	No	Yes	Yes
View Favorites Tab	No	Yes	Yes
View My Listings (Dashboard)	Yes	No	No
View My Inquiries (Dashboard)	No	Yes	Yes
View My Transactions (Dashboard)	No	Yes	Yes

5.3 Navigation

The navigation bar in `header.php` adapts based on the session. Logged-out users see Login and Register links. Buyers and renters see a Favorites tab. Agents do not see the Favorites tab. All users see Dashboard and Logout when logged in.

6. Challenges and Solutions

Challenge 1: MySQL Port Conflict

When starting XAMPP, MySQL failed to start with the error: MySQL shutdown unexpectedly. The cause was that port 3306 was already in use by another process running on the machine.

Solution: The MySQL port in XAMPP was changed from 3306 to 3307 by editing the my.ini configuration file. The DB_HOST constant in config/config.php was then updated to localhost:3307 to match the new port.

Challenge 2: Sample User Password Hashes

After populating the database with sample users, logging in as agent_maria, buyer_james, or renter_lisa failed even with the correct password. The placeholder hashes in the original SQL script were not valid bcrypt hashes and could not be verified by password_verify().

Solution: A corrected SQL UPDATE statement was run in MySQL Workbench to replace the placeholder hashes with a valid bcrypt hash generated from the password: password. All three sample accounts then logged in successfully.

Challenge 3: Database Connection Error After Port Change

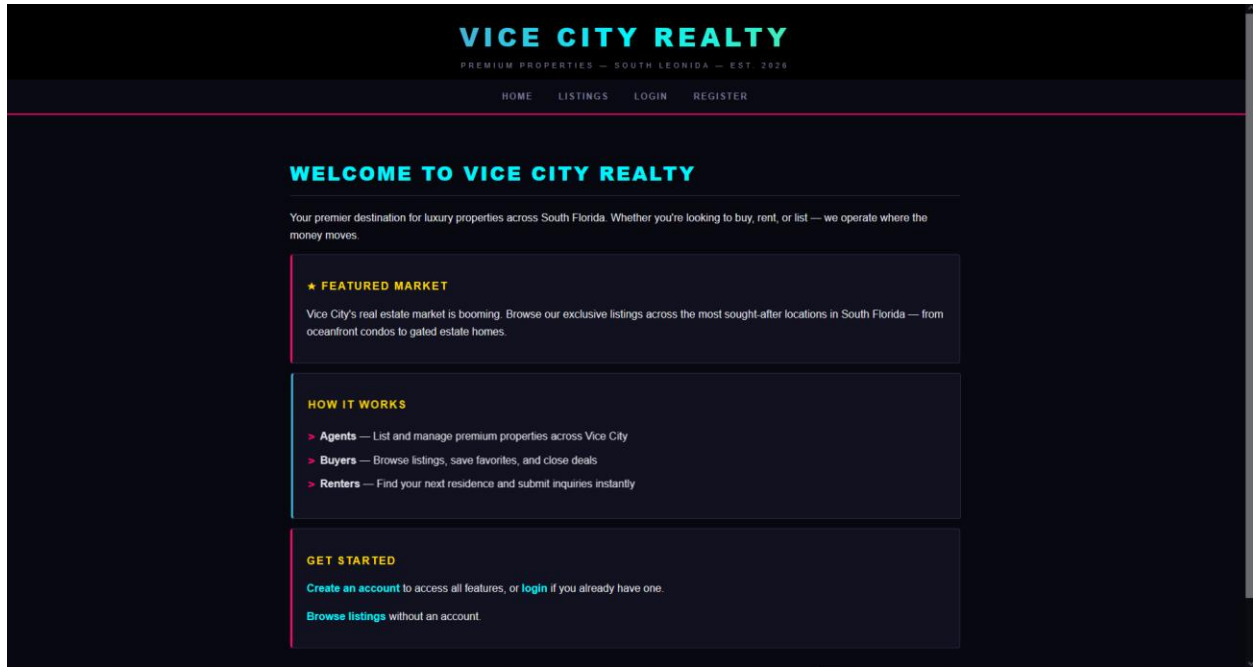
After changing the MySQL port in XAMPP, the PHP application threw a fatal error: Database connection failed: SQLSTATE[HY000][2002] No connection could be made because the target machine actively refused it.

Solution: The DB_HOST value in config/config.php was updated from localhost to localhost:3307 to point PHP's PDO connection to the correct port. The error was immediately resolved after saving the file.

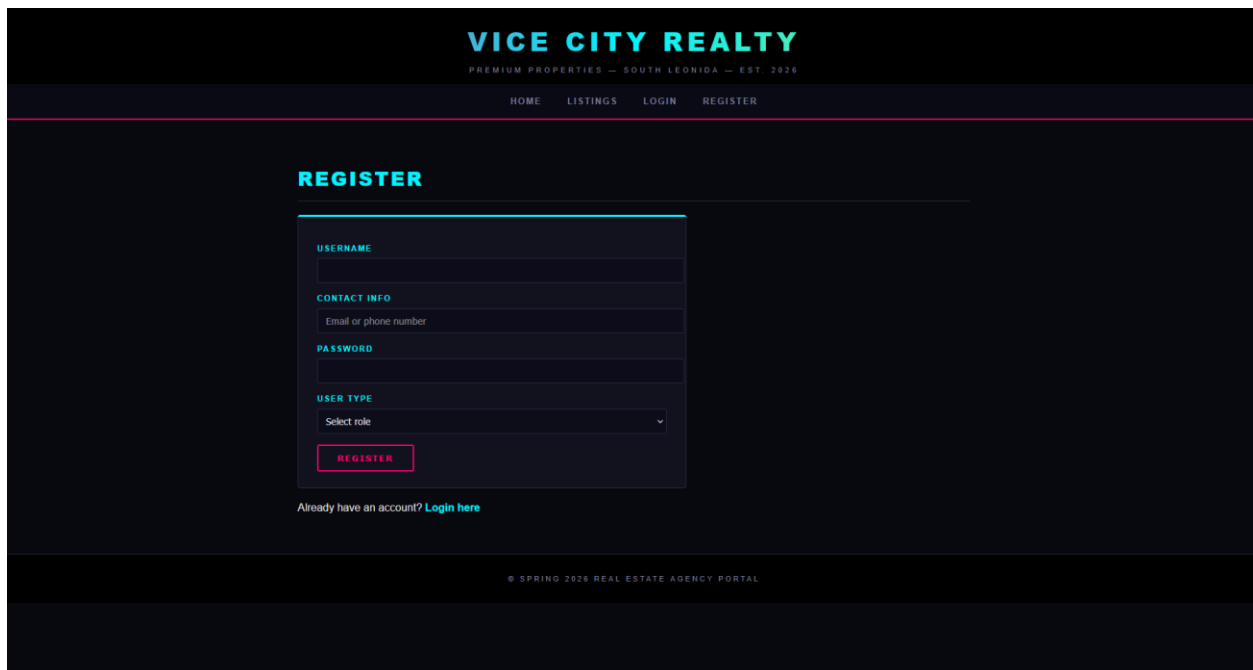
7. Screenshots

The following screenshots demonstrate the working platform. Insert your screenshots below each label.

Homepage



Registration Page



Login Page

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH FLORIDA — EST. 2024

HOME LISTINGS LOGIN REGISTER

LOGIN

USERNAME

PASSWORD

LOGIN

Don't have an account? [Register here](#)

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Agent Dashboard

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2028

[HOME](#) [LISTINGS](#) [ADD LISTING](#) [DASHBOARD](#) [LOGOUT](#)

DASHBOARD

Welcome: agent_maria

Role: agent

Contact: maria@agency.com

AGENT ACTIONS

[Add New Property](#)

MY LISTINGS

Coconut Grove

Vice City — \$200,000.00

Status: available

[View](#) [Edit](#)

Wynwood Modern Apartment

South Leonida — \$1,800.00

Status: available

[View](#) [Edit](#)

Leonida Beach Home

Key Biscayne — \$820,000.00

Status: sold

[View](#) [Edit](#)

Downtown Studio Apartment

Financial District — \$2,200.00

Status: rented

[View](#) [Edit](#)

Luxury Leonida Condo

Downtown Vice City — \$850,000.00

Status: sold

[View](#) [Edit](#)

BROWSE

[View All Properties](#)

Add Property

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME LISTINGS ADD LISTING DASHBOARD LOGOUT

ADD PROPERTY

TITLE

PROPERTY TYPE

Apartment, House, Condo...

ADDRESS

CITY

PRICE

STATUS

available

ADD PROPERTY

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Edit Property

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME LISTINGS ADD LISTING DASHBOARD LOGOUT

EDIT PROPERTY

TITLE

Coconut Grove

PROPERTY TYPE

Loft

ADDRESS

1750 Amsterdam Avenue

CITY

Vice City

PRICE

200000.00

STATUS

Available

SAVE CHANGES Cancel

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Property Listings

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME LISTINGS ADD LISTING DASHBOARD LOGOUT

PROPERTY LISTINGS

BEACH SIDE APARTMENT

Type: Condo

Address: 284 Ocean Drive

City: Vice City

Price: \$500000.00

Status: available

Agent: new_agent

View Details

COCONUT GROVE

Type: Loft

Address: 1750 Amsterdam Avenue

City: Vice City

Price: \$200000.00

Status: available

Agent: agent_maria

Property Details

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME LISTINGS ADD LISTING DASHBOARD LOGOUT

PROPERTY DETAILS

BEACH SIDE APARTMENT

Type: Condo

Address: 284 Ocean Drive

City: Vice City

Price: \$500,000.00

Status: available

Agent: new_agent

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Submit Inquiry

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME

LISTINGS

FAVORITES

DASHBOARD

LOGOUT

SUBMIT INQUIRY

MESSAGE

SEND INQUIRY

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Favorites Page

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME

LISTINGS

FAVORITES

DASHBOARD

LOGOUT

MY FAVORITES

BEACH SIDE APARTMENT

City: Vice City

Price: \$500,000.00

Status: available

Agent: new_agent

Saved On: 2026-05-27 08:12:41

View Details

REMOVE

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Buyer/Renter Dashboard

VICE CITY REALTY

PREMIUM PROPERTIES — SOUTH LEONIDA — EST. 2026

HOME

LISTINGS

FAVORITES

DASHBOARD

LOGOUT

DASHBOARD

Welcome: buyernew

Role: buyer

Contact: 1234

QUICK ACTIONS

[Browse Properties](#) [My Favorites](#)

MY FAVORITES

Beach Side Apartment

Vice City — \$500,000.00

Status: available

[View](#)

MY INQUIRIES

Tables

	Table	Action	Rows	Type	Collation	Size	Overhead
☐	favorites	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	inquiries	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	properties	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐	transactions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	users	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
	5 tables	Sum	0	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

8. Conclusion

The Real Estate Agency Portal successfully demonstrates a fully functional PHP and MySQL web application. The platform supports three distinct user roles with enforced access controls, secure authentication using bcrypt password hashing, and a normalized relational database with stored procedures, a view, and an automatic trigger.

The project covers all required features including property management by agents, inquiry submission by buyers and renters, a favorites system, a role-specific dashboard, and a complete transaction recording system. All forms connect to the database using PDO prepared statements to prevent SQL injection.

Through building this project, I gained practical experience was gained in PHP session management, object-oriented database abstraction, MySQL relational design, and deploying a local web application using XAMPP.